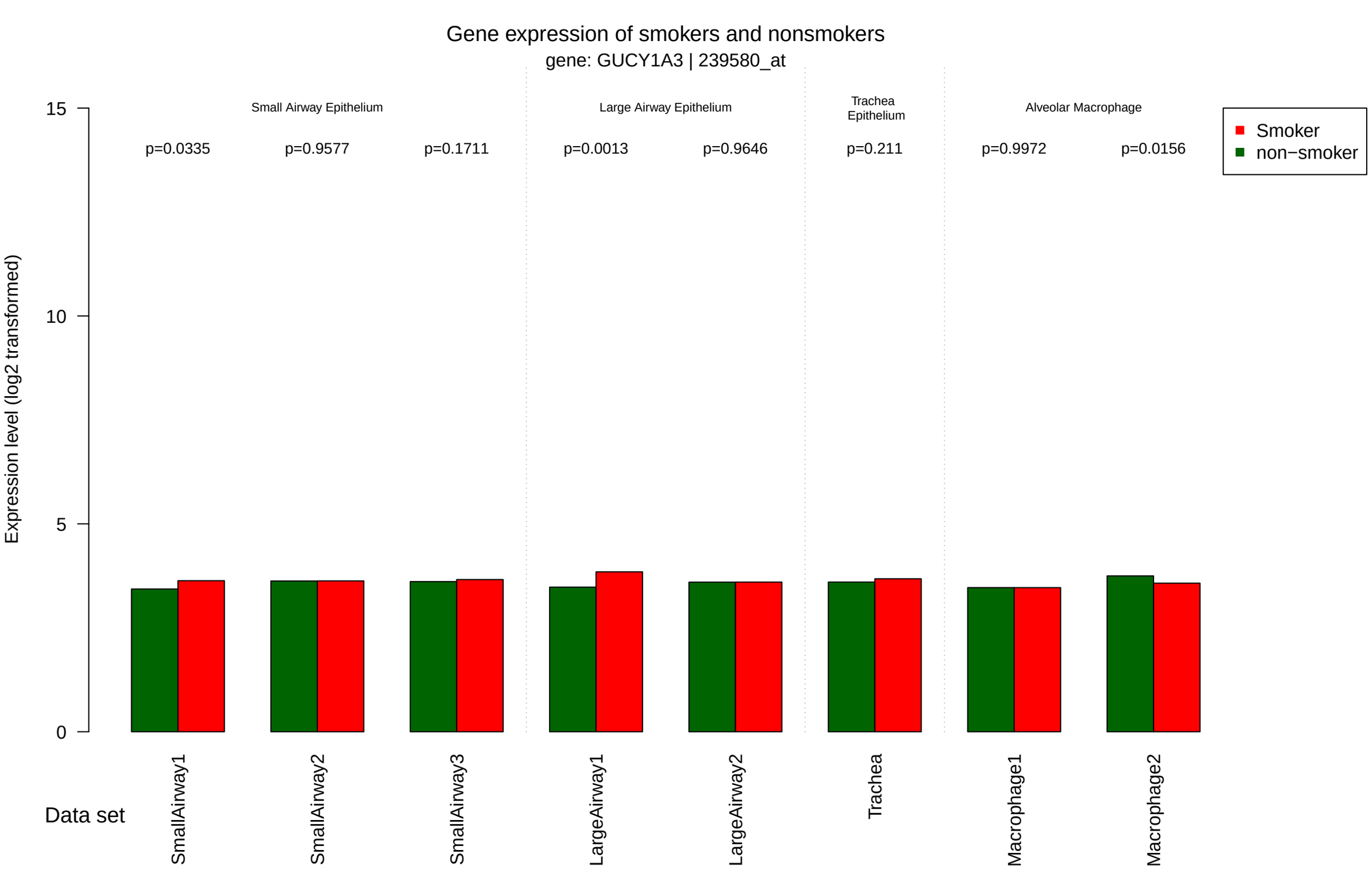
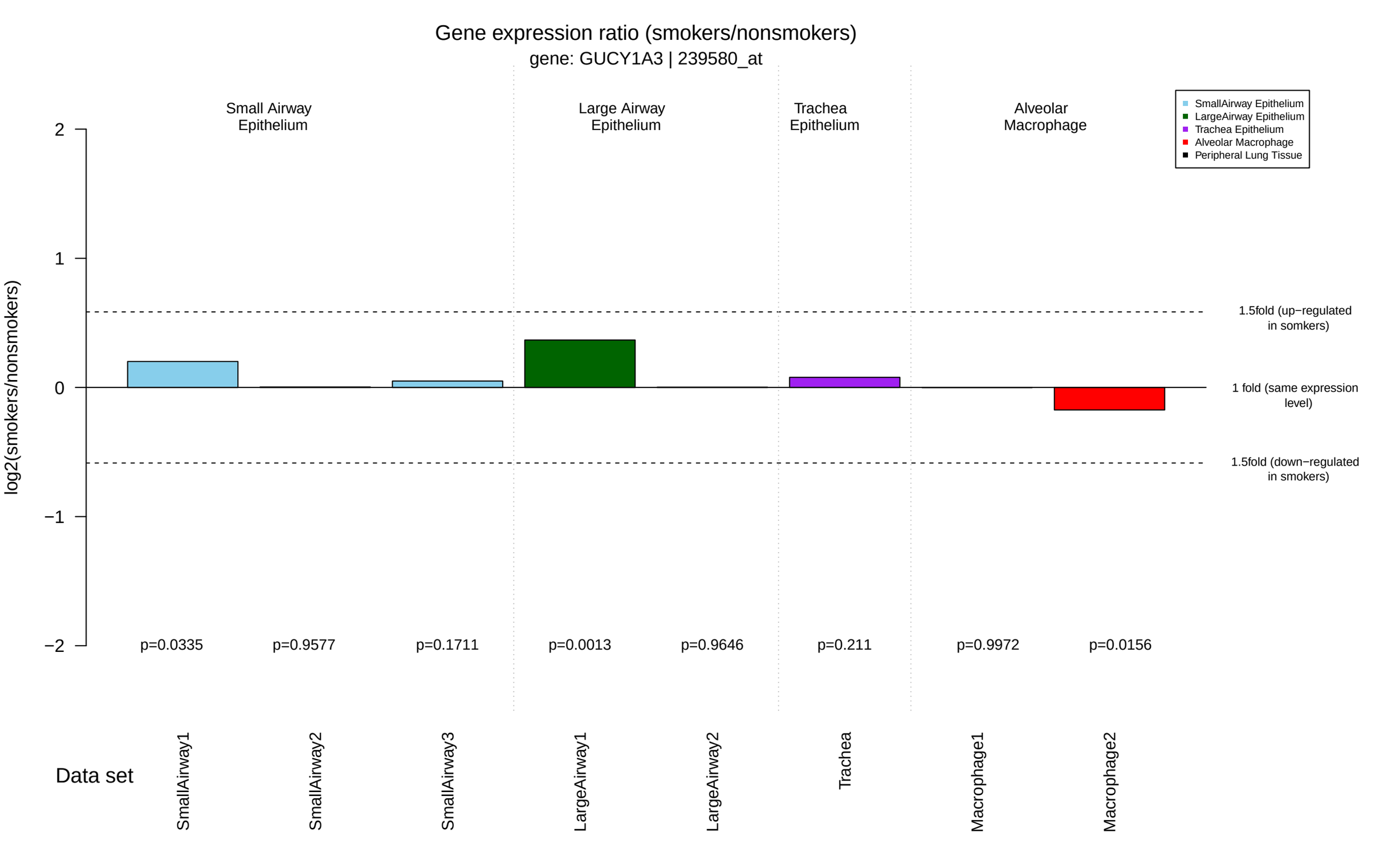
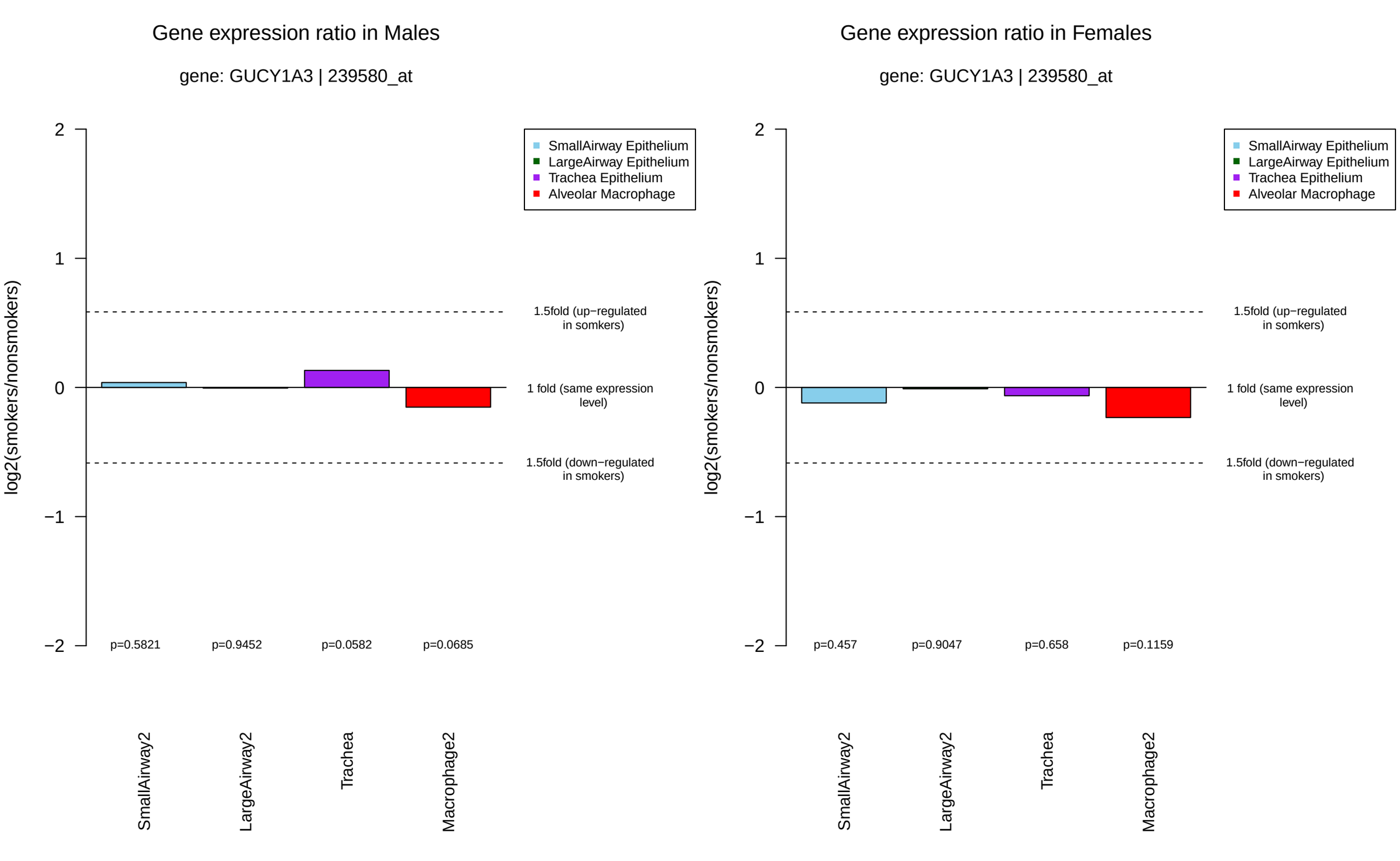
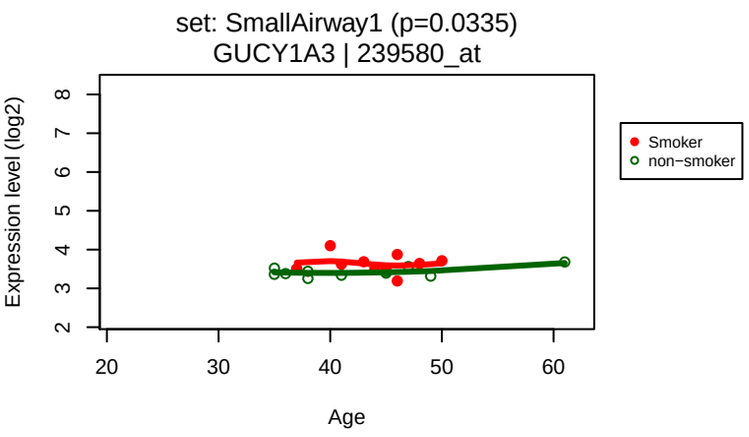
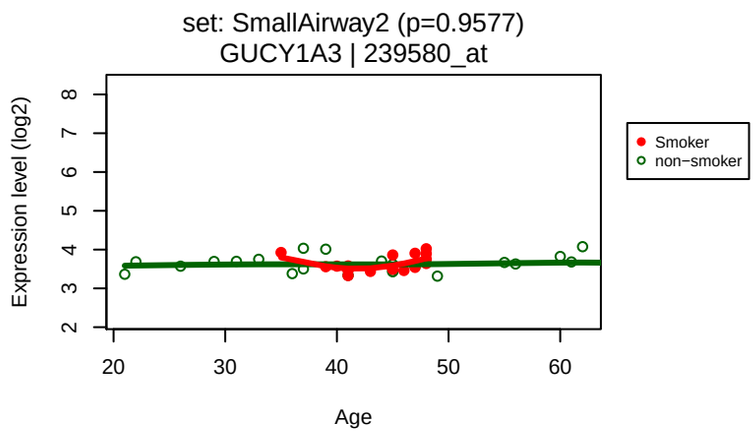
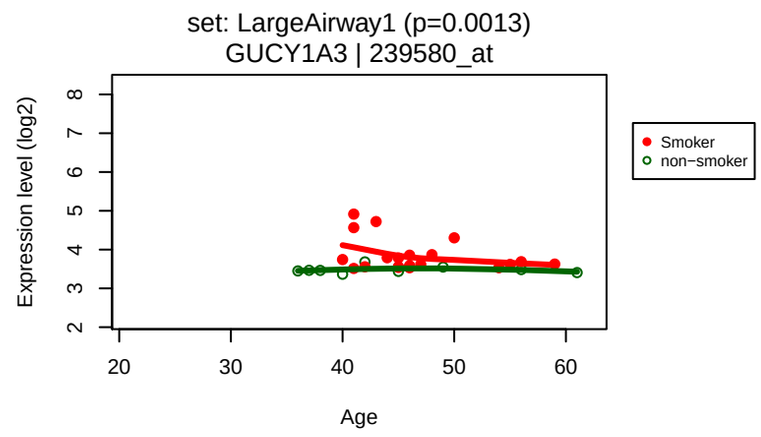

Gene expression of smokers and nonsmokers
gene: GUCY1A3 | 239580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0335
p=0.9577
p=0.1711
p=0.0013
p=0.9646
p=0.211
p=0.9972
p=0.0156
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

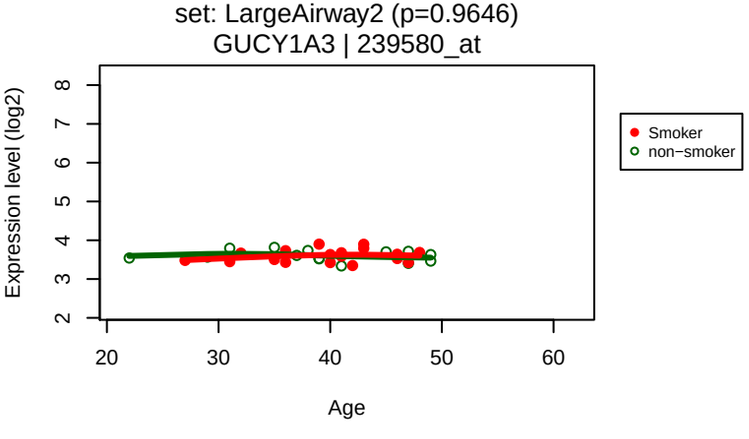

Gene expression ratio (smokers/nonsmokers)
gene: GUCY1A3 | 239580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0335
p=0.9577
p=0.1711
p=0.0013
p=0.9646
p=0.211
p=0.9972
p=0.0156
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

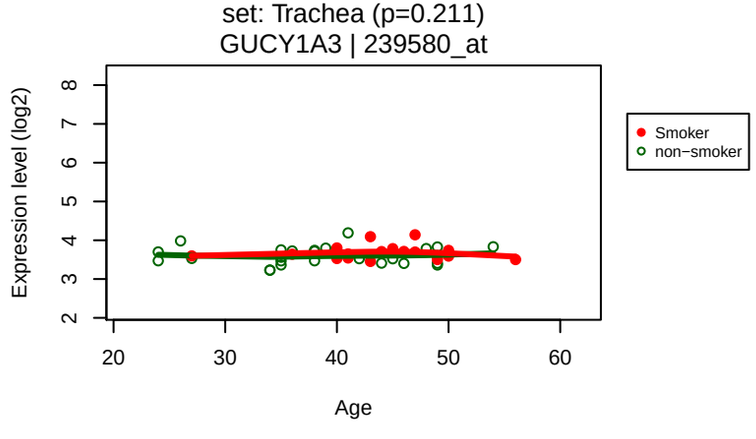

Gene expression ratio in Males
gene: GUCY1A3 | 239580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5821
p=0.9452
p=0.0582
p=0.0685
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GUCY1A3 | 239580_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.457
p=0.9047
p=0.658
p=0.1159
SmallAirway2
LargeAirway2
Trachea
Macrophage2

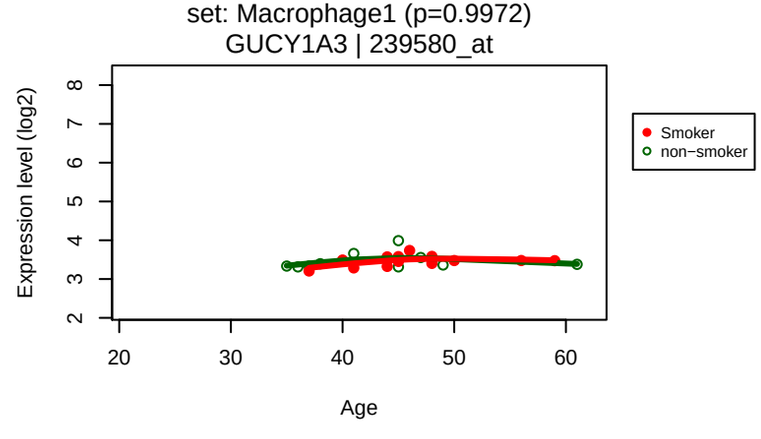

set: SmallAirway1 (p=0.0335)
GUCY1A3 | 239580_at
Expression level (log2)
Age
Smoker
non−smoker

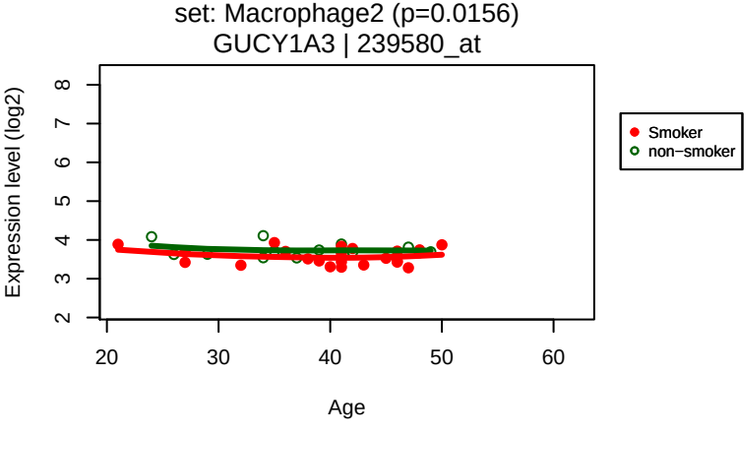

set: SmallAirway2 (p=0.9577)
GUCY1A3 | 239580_at
Expression level (log2)
Age
Smoker
non−smoker

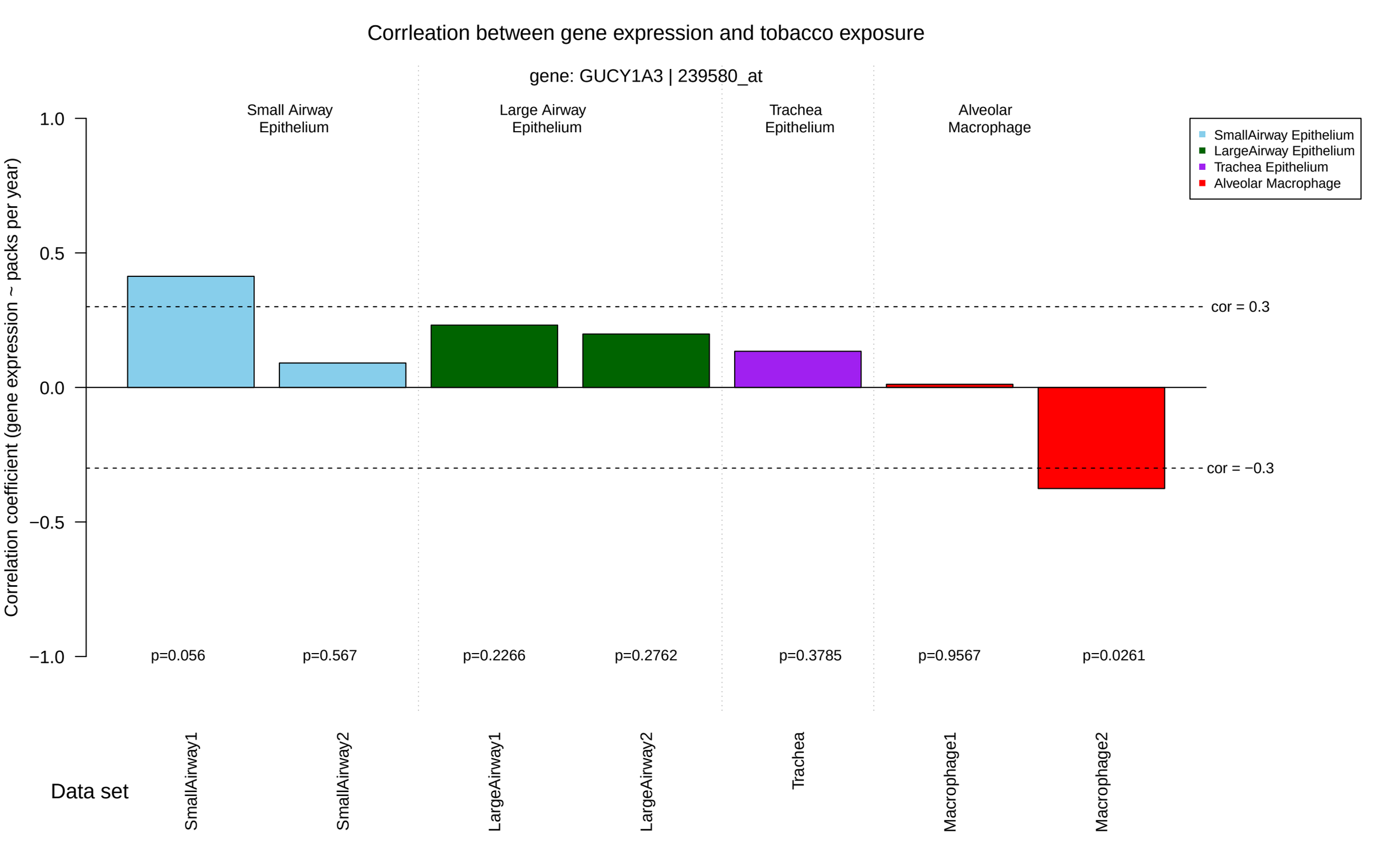

set: LargeAirway1 (p=0.0013)
GUCY1A3 | 239580_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9646)
GUCY1A3 | 239580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.211)
GUCY1A3 | 239580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9972)
GUCY1A3 | 239580_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0156)
GUCY1A3 | 239580_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GUCY1A3 | 239580_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.056
p=0.567
p=0.2266
p=0.2762
p=0.3785
p=0.9567
p=0.0261
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2